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Case Competition 2025-26

RTM: A Comprehensive, Context-Aware Solution to Prevent Crime

Broke Street Bets

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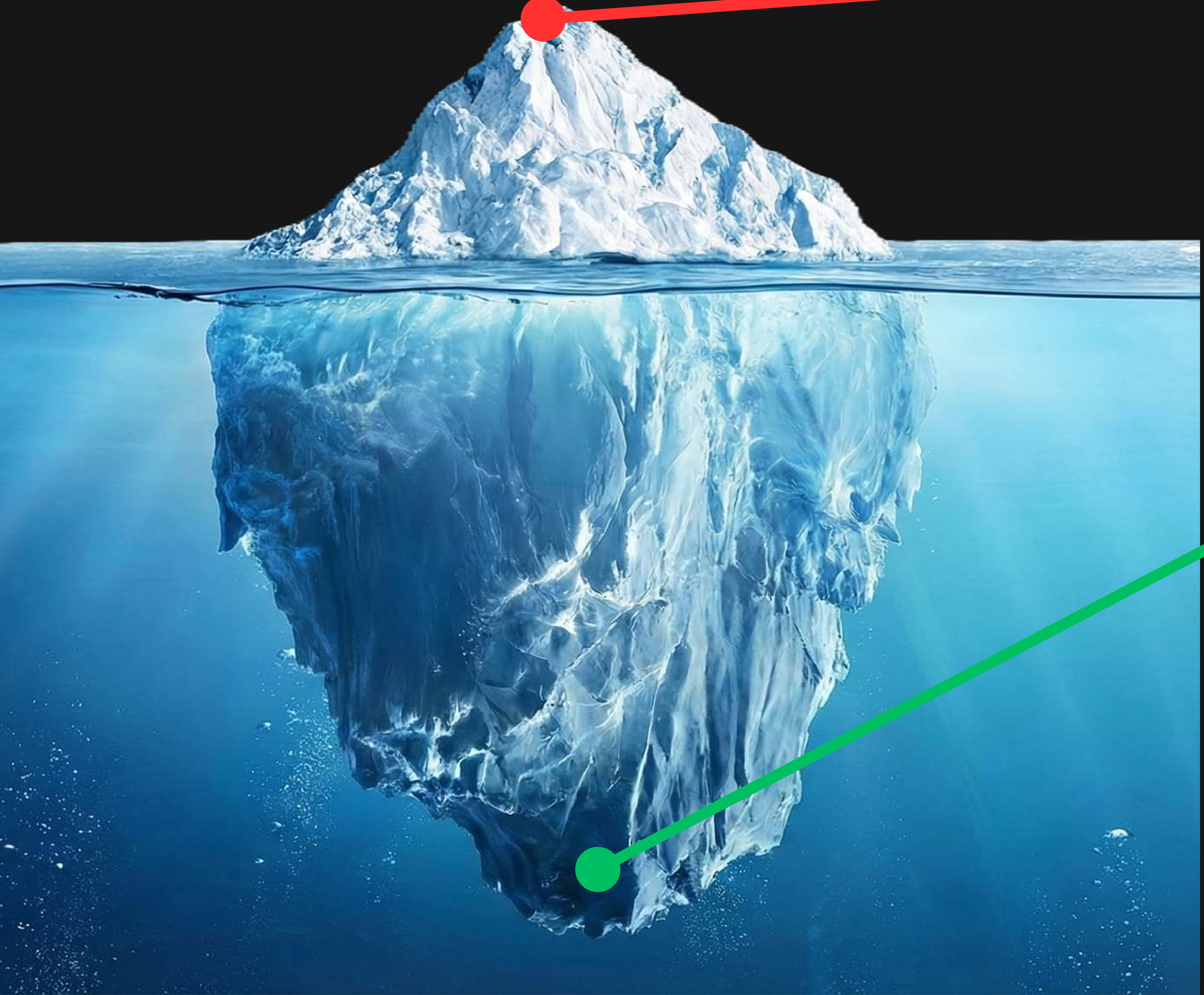
Most crime solutions focus on the tip, we want to focus on the root



Managing Damage

We spend most of our resources on Post facto crime: Trying to catch criminals. Creating better catching techniques still counts as fighting the sympotoms and not the disease.

Managing Risk

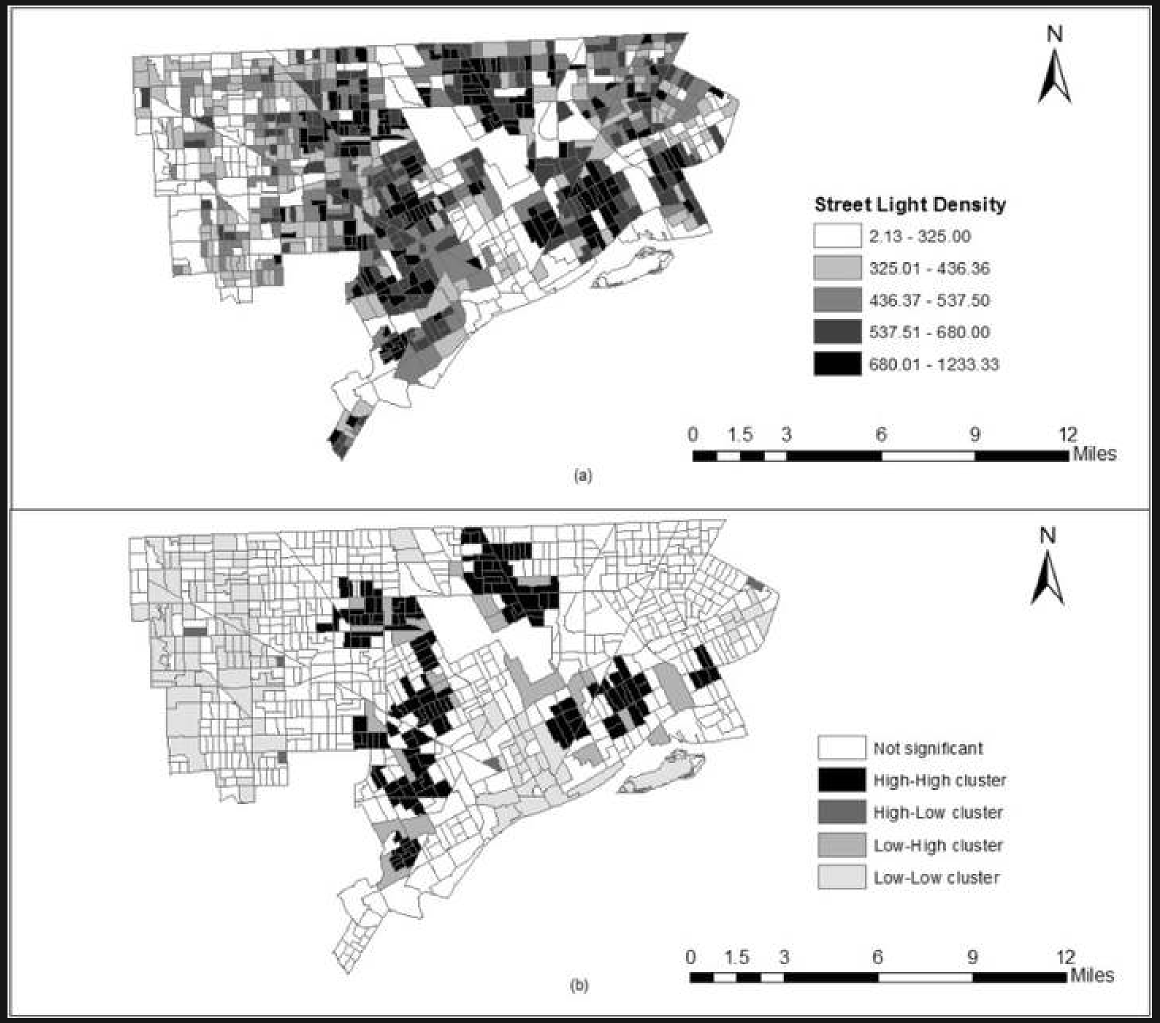
Developed nations prioritize Pre-emption: Neutralizing the opportunity for crime before it starts. India completely lacks this predictive layer, leaving the root causes unaddressed while we wait for the crime to happen

What causes crime?

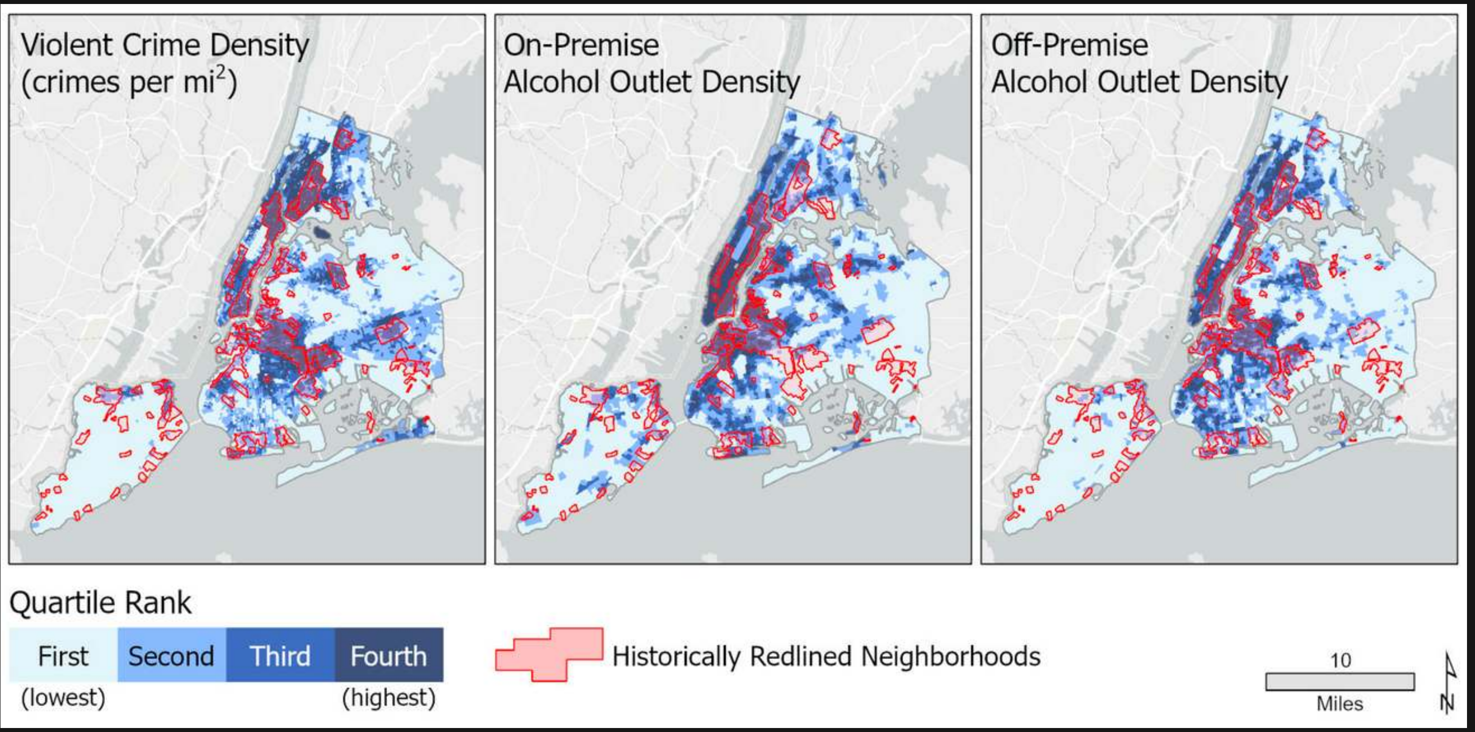
We need to go from who did it to what caused it

Environmental Triggers & Aggression Nodes (The Alcohol): Violent crime density is 4x higher within 500 meters of high-density alcohol outlets compared to city averages, creating predictable "aggression nodes." This correlation allows us to target static environmental triggers rather than chasing dynamic human behavior.

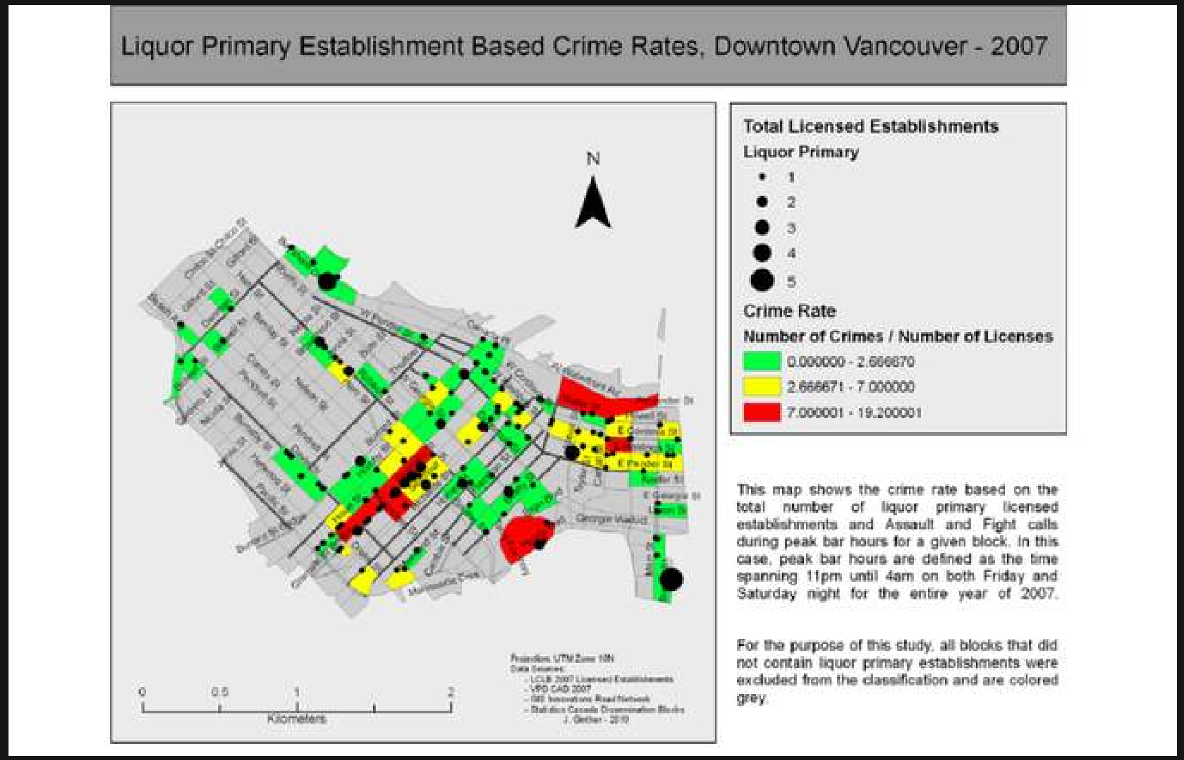
Darkness: Global meta-analysis confirms that improved street lighting reduces crime by 21% in experimental areas, while specific audits in Indian cities (e.g., Delhi) reveal that 68% of street crimes occur in "Dark Spots" (<5 Lux). Improved street lighting reduced crime by 21% in experimental areas. (Source given below)



The impact of street lights on spatial-temporal patterns of crime in Detroit, Michigan, Cities Volume 79, 2018, Pages 45-52, ISSN 0264-2751



There is a high co relation between crime and environmental factors like proximity to liquor shops, income and lighting etc



(Source: Welsh & Farrington, "Surveillance & Lighting," Campbell Systematic Reviews, 2022; Safetipin Safety Assessment Report, 2019)

Bars on blocks: A cellular automata model of crime and liquor licensed establishment density, September 2012, Computers Environment and Urban Systems 36(5):412–422 DOI:10.1016/j.compenvurbsys.2012.02.004

Bharatiya Risk Terrain Modeling: The Solution to a Nuanced and Context-Aware Response to Crime

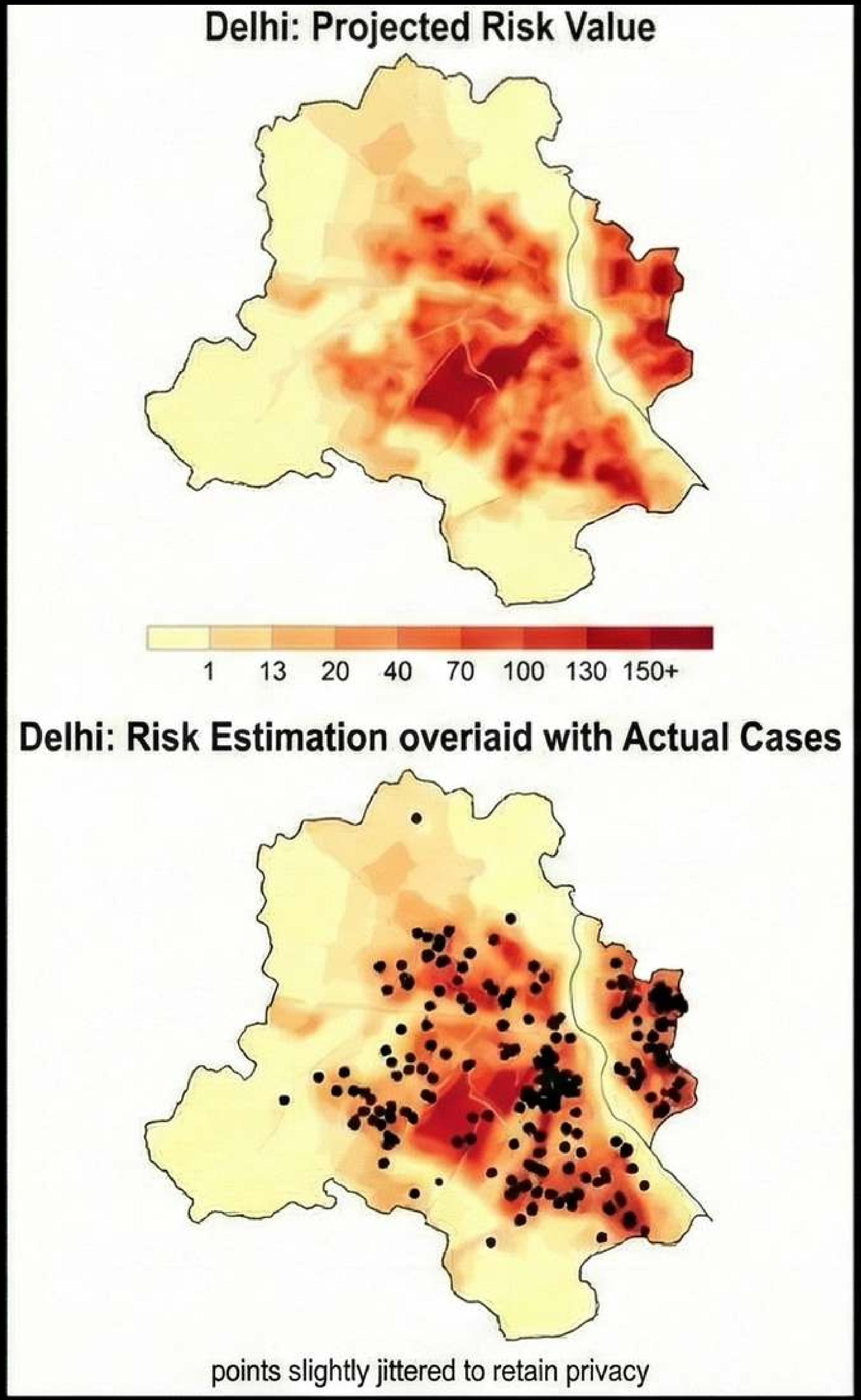
What is It?

- A Multi-Dimensional Diagnostic Tool: A sophisticated analytical framework that maps environmental, social, and economic determinants to identify "risk terrains" where crime is most likely to emerge.
- The Convergence of Risk Factors: It visualizes how various situational factors (e.g., infrastructure, abandoned spaces, or socio-economic stressors) overlap to create opportunities for crime.
- Community-Centric Prevention: An inclusive approach that integrates community-specific data to move beyond traditional policing toward holistic crime prevention.

How does it Help?

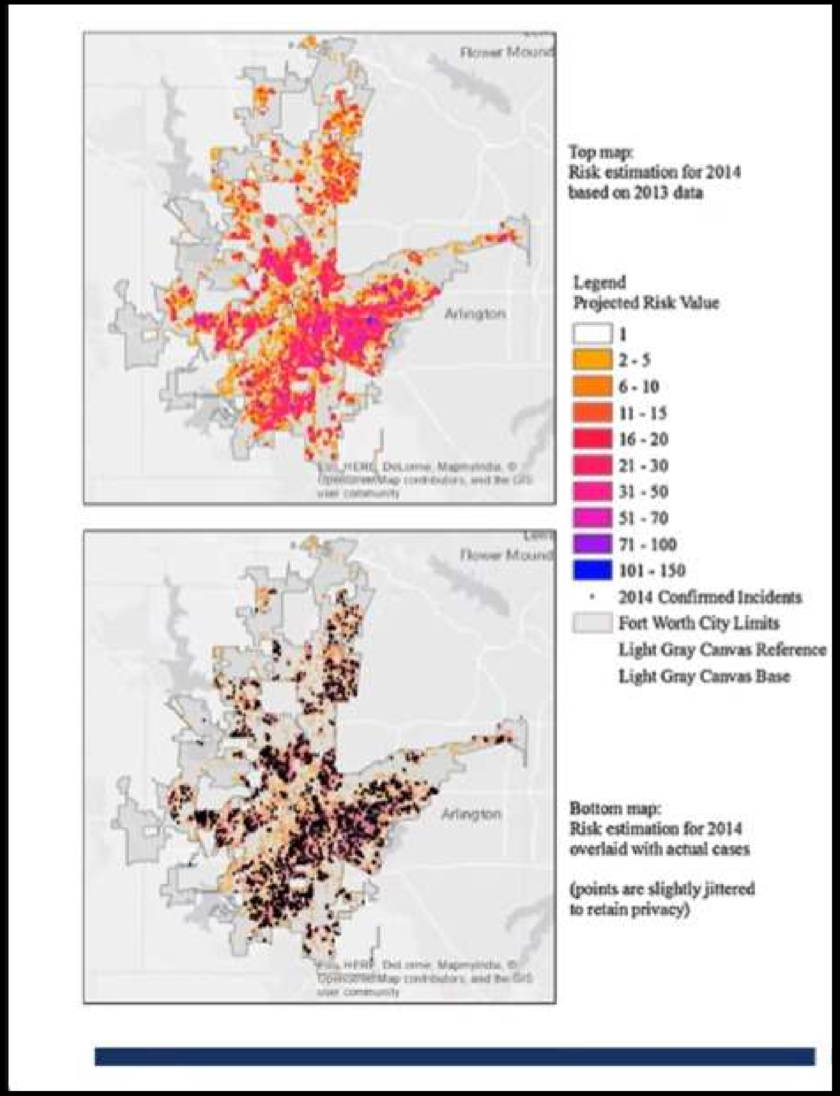
- Precision-Led Policing: Enables law enforcement to accurately forecast high-risk locations, allowing for smarter patrolling and more effective preventive measures.
- Community Outreach & Support: Identifies vulnerable areas to facilitate pre-emptive social support and create better opportunities for individuals at risk of falling into criminal activity.
- Proactive Resource Management: Significantly reduces the post-facto burden on police by preventing crimes before they occur, rather than simply reacting to them.

BRTM: The Delhi Experiment



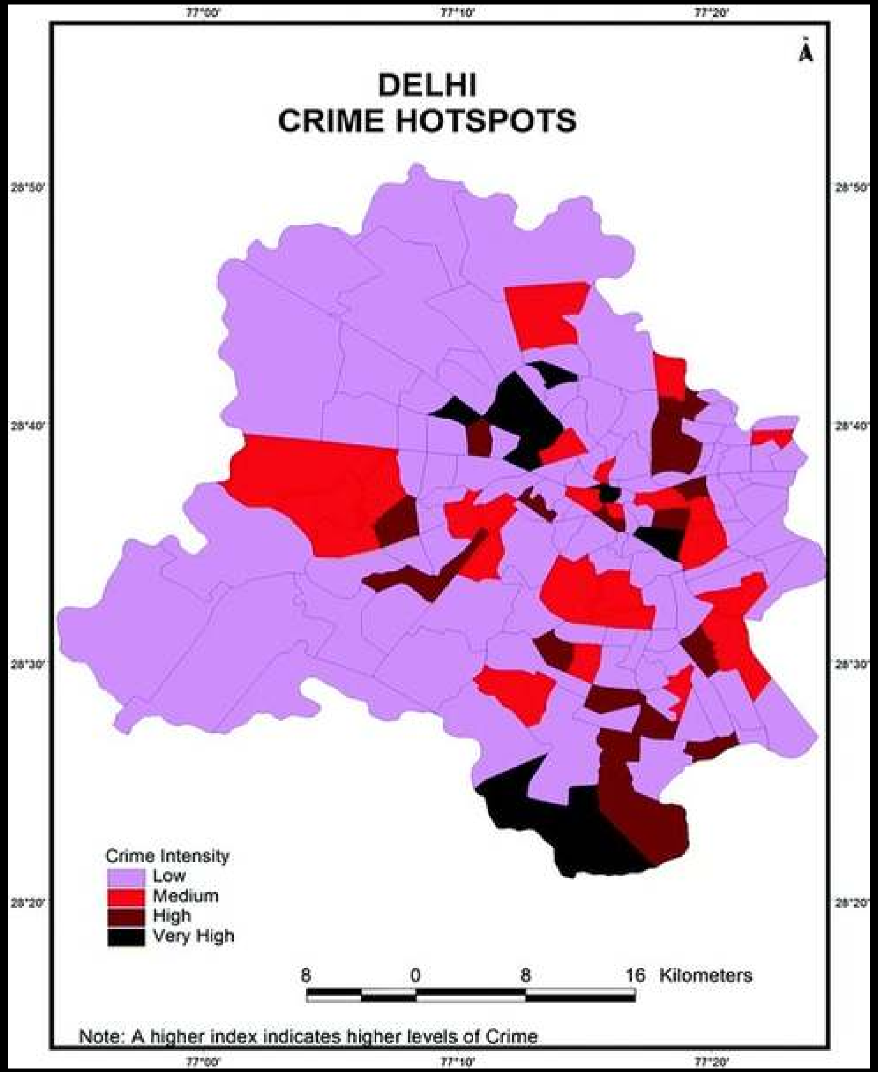
*Rough data from Hindustan times 2015, Mapping crimes reported in June

- We deployed the BRTM model in the context of Delhi.
- Divided the city into 500m by 500m blocks
- Used a multi-dimensional dataset including Income, Sanitation, Water Availability, Lighting, Intoxicant Availaibility etc.
- Findings:
- Poverty and infrastructural deficiencies were the most influential factors.
- 55% of future incidents were predicted from 10% of highest risk blocks.



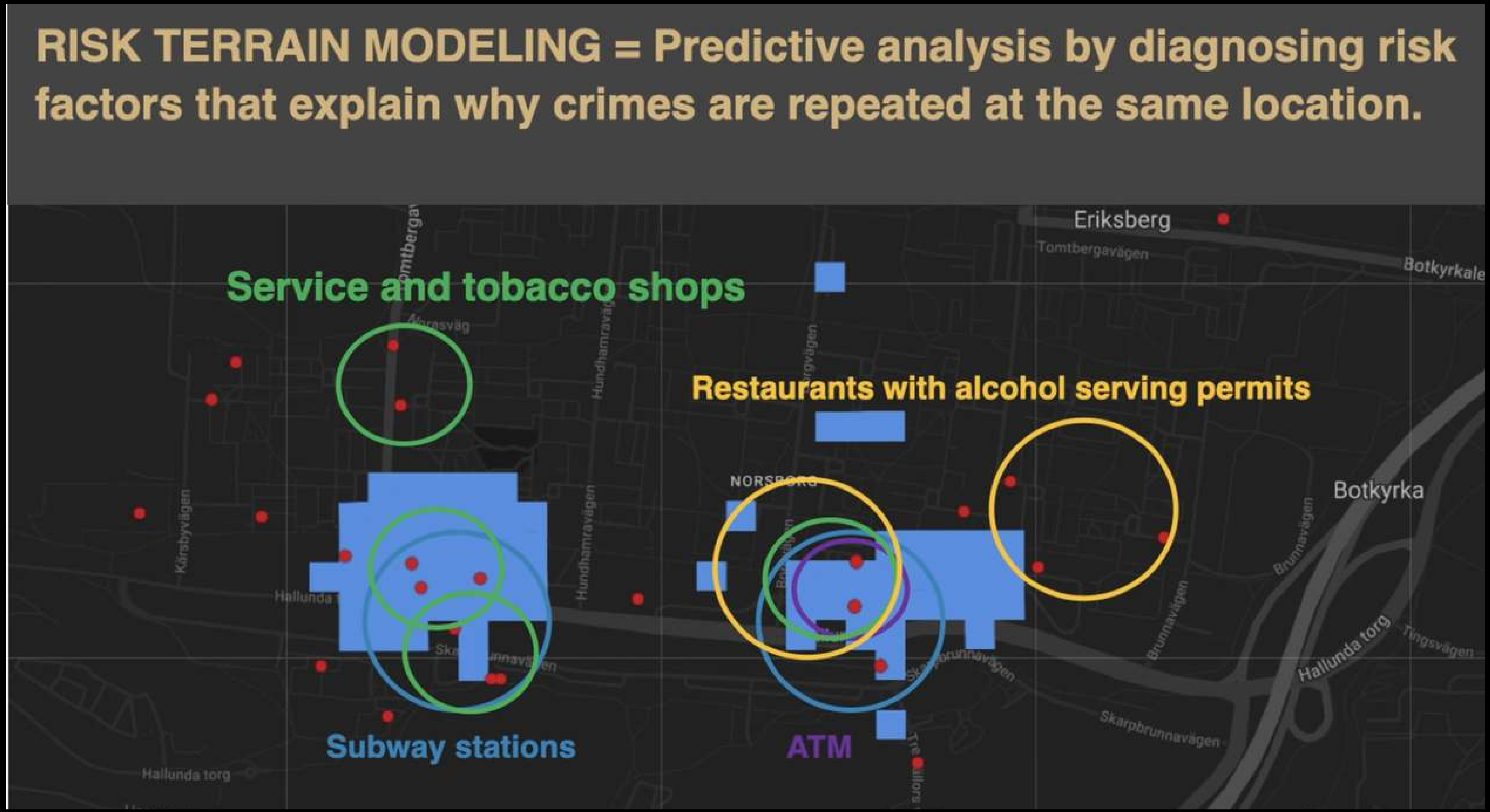
Our approach is taken from an equivalent analysis on texas

We ran our software pilot on a very small part of Delhi with very less factors, a bigger implementation at scale can be highly useful in crime prediction with higher accuracy.



REACTIVE

PREDICTIVE



Old way of hotspot mapping (eg. Delhi/Telangana Police)

Ineffective since it is only based on previous criminal data

New way: RTM (used by Swedish police, piloted in New Jersey)

RTM layers data on subways, ATMs, and shops to identify environmental risks and forecast crime locations before incidents occur.

Hotspot mapping assumes criminals stay in one place. RTM assumes criminals seek specific environmental conditions (darkness, isolation, lack of surveillance). By mapping the conditions, we intercept the crime before the red dot ever appears.

Systems Integration & Data Flow Architecture

Integration Sources

System Source	Method	Function
Municipal Light	API / Sync	Monitor
Police Logs	Batch + Live	Awareness
Traffic Jnc	Geospatial	Mapping
Event-Driven	Trigger API	Coord.
PCR Dispatch	Priority Q	Response

Total Systems: 5 — Uptime: 99.9%

Data Flow Pipeline

01 INPUTS

- **1,400** Junction Records
- **5,000** Lighting Updates
- **8,000** Historical Pts

Volume: 10-15 GB Raw Data

02 PROCESSING

- Spatial Clustering
- Risk Scoring Algo

Freq: **Weekly** — < 30m

03 OUTPUTS

150
HIGH RISK LOCATIONS
Top 10% Identified

Score Range	Action
0-100	Ranked

24-Month Project Execution Roadmap

Active Phase

MONTHS 1-8	MONTHS 8-10	MONTHS 10-18	MONTHS 18-21	MONTHS 21-24
Approval	Resource	Develop & API	Testing	Review

Project Kickoff — Phase 1 Gate — Deployment — Handover

Taking Delhi as an example

PHASE 1 – PLANNING & DATA COLLECTION

(Months 0-6 | Build the risk foundation)

Objective: Create a **city-wide risk map** by identifying physical and environmental factors that consistently correlate with crime.

What data needs to be colle be collected :

Data Layer	Scale (Delhi Example)	Source
Liquor shops & bars	-1,200 outlets	Excise Dept
Poorly lit streets / zones	-2,000 micro-areas	Municipal + night audits
Major junctions / alleys	-1,500 nodes	Traffic Police / PWD
Transit hubs (metro, bus)	-500	DMRC / DTC
Historical incidents (3 yrs)	-30,000 records	Media records

1. **Data consolidation (2 months)**
   - Merge GIS layers
   - Standardise formats
   - Remove duplicates
2. **Ground validation (2 months)**
   - Sample night-time lighting surveys
   - Validate blind spots, alley access, visibility gaps
3. **Risk scoring (2 months)**
   - Each location assigned a **composite risk score**
   - Output: ranked list of high-risk micro-locations

PHASE 2 – PREDICTION & SYSTEM ADOPTION

(Months 6-14 | Turn insight into decisions)

Objective : Convert consolidated city risk data into a predictive prioritisation system and ensure police adoption at the operational level.

What Is Implemented:

- Predictive Risk Prioritisation System City: Locations ranked monthly based on risk concentration Focus on top 10-15% high-risk micro-locations (-150 sites) System explains why a location is high-risk (lighting, land-use, access) What the current system does NOT do :
  - No personal data tracking
  - No replacement on police judgement
  - The system supports decisions – it does not automate enforcement.
- Operational Dashboard for Police & City Officials Simple map + ranked list view District-wise filtering Before-after comparison to track improvements Designed for Police planners Traffic & municipal officers Command-level decision-makers Built for usability, not technical complexity.
- Police Training & Change Enablement 1-2 day training workshops per district Clear Standard Operating Procedures (SOPs): "If a location enters the top risk tier → action within X days" Focus on interpretation, not technology Outcome Faster planning cycles Uniform decision-making across districts

PHASE 3 – EXECUTION & DIRECT IMPACT

(Months 14-24 | Prevent escalation on the ground)

Objective: Act only where risk is proven and measure impact. What actually happens

- Targeted Physical Interventions :

Intervention Type	Avg. Cost per Location	Approx. Count
Lighting repair / upgrade	₹2-3 lakh	-1000
Visibility & signal tweaks	₹1-2 lakh	-40
Minor street redesign	₹5-7 lakh	-10

- Patrol & Response Optimization :

Patrol routes updated monthly, Faster response in predicted hotspots, No additional manpower Measured Impact (conservative)

Metric	Before	After RTM
Average response time	12-15 minutes	7-9 minutes
Repeat incidents at hotspots	Baseline	↓ 20-30%
Blind-spot related crimes	Baseline	↓ -25%

Phase-1

Total estimated cost:6-8 cr:

Cost Head	Units / Assumption	Cost (₹ Cr)	Explanation
GIS Data Integration	2 GIS analysts × 6 months	1.2	Merging junction maps, land-use layers, street layouts
Urban Risk Analysis	2 planners × 6 months	1.0	Identifying risk factors (lighting, liquor proximity, alleys)
Field Lighting Audits	4 teams × 8 weeks	2.2	Night surveys to validate poorly lit micro-areas
Data Cleaning & Validation	2 data engineers × 4 months	1.1	Removing noise, normalising datasets
Inter-agency Coordination	PMO + legal + access	0.8	Data permissions, coordination with police/municipality
Contingency (≈10%)	–	0.7	Field variability, resurvey needs

Phase-2

Total estimated cost:8-10 cr:

Cost Head	Units / Assumption	Cost (₹ Cr)	Explanation
Predictive System Build	2 data scientists × 8 months	2.0	Risk scoring, prioritisation logic
Backend & Data Pipelines	2 engineers × 8 months	1.8	Automated refresh, secure data flow
Operational Dashboard	1 dev + UI support	1.0	Simple map & ranked lists for police
System Testing & QA	3 months	0.8	Validation, stress testing
Police Training	1–2 day workshops × districts	1.4	SOPs, interpretation, adoption
System Ops (12 months)	Cloud + maintenance	1.5	Low compute, monitoring
Contingency (≈10%)	–	0.9	Iterations, refinements

Phase-3

Total estimated cost: 30-32cr

Intervention	Unit Cost	Count	Cost (₹ Cr)	Explanation
Lighting Repair / Upgrade	₹2.5 lakh	1000	25	Fixing dark spots, visibility
Signal / Junction Tweaks	₹1.5 lakh	40	0.6	Timing, visibility, signage
Minor Street Redesign	₹6.0 lakh	10	0.6	Barriers, sightlines
Maintenance (1 year)	–	–	3.0	Upkeep of upgrades
Patrol Route Optimisation	Process change	–	0.5	No manpower increase
Project Management	–	–	1.2	Execution coordination
Contingency	–	–	2.0	Ground realities

Thank You!